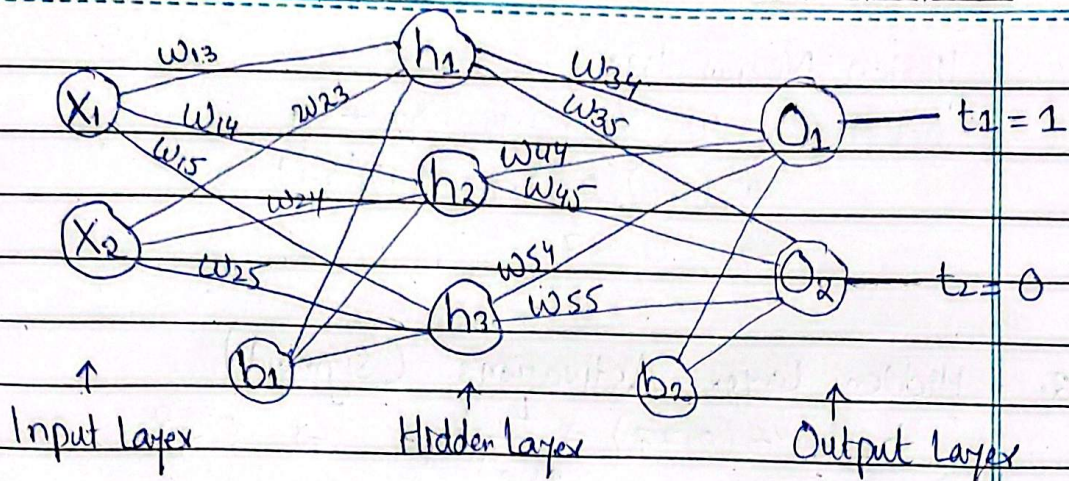


ML - Assignment : 02

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Given Values:-

Inputs: $X_1 = 0.6$, $X_2 = 0.1$

Targets: $t_1 = 1$, $t_2 = 0$

$\eta = 0.1$

Initial Weight Initialization:-

$w_{13} = 0.4$, $w_{14} = 0.2$, $w_{15} = 0.6$

$w_{23} = 0.3$, $w_{24} = 0.5$, $w_{25} = 0.1$

$b_1 = 5$, $b_2 = 5$

$w_{34} = 0.7$, $w_{35} = 0.3$, $w_{44} = 0.4$

$w_{45} = 0.6$, $w_{54} = 0.8$, $w_{55} = 0.2$

STEP: 01 Forward Propagation:-

1. Hidden Layer Net Inputs

Hidden Neuron h_1 :-

$$\text{net}_{h_1} \Rightarrow w_{13} \cdot X_1 + w_{23} \cdot X_2 + b_1$$

$$\Rightarrow (0.4)(0.6) + (0.3)(0.1) + 0.5$$

$$\Rightarrow 0.77$$

Hidden Neuron h_2 :-

$$\text{net}_{h_2} \Rightarrow w_{14} \cdot X_1 + w_{24} \cdot X_2 + b_1$$

$$\Rightarrow (0.2)(0.6) + (0.5)(0.1) + 0.5$$

$$\Rightarrow 0.67$$

Hidden Neuron h_3 :

$$\text{net-}h_3 \Rightarrow w_{15} \cdot x_1 + w_{25} \cdot x_2 + b_1$$

$$\Rightarrow (0.6)(0.6) + (0.1)(0.1) + 0.5$$

$$\Rightarrow 0.87$$

2. Hidden Layer Activations (Sigmoid)

$$h_1 = \sigma(0.77) \Rightarrow \frac{1}{1+e^{-0.77}} \Rightarrow 0.68$$

$$h_2 = \frac{1}{1+e^{-0.67}} \Rightarrow 0.66$$

$$h_3 = \frac{1}{1+e^{-0.87}} \Rightarrow 0.70$$

3. Output Layer Net Inputs:

Output Neuron O_1 :

$$\text{net-}O_1 = w_{34} \cdot h_1 + w_{44} \cdot h_2 + w_{54} \cdot h_3 + b_2$$

$$\Rightarrow (0.7)(0.68) + (0.4)(0.66) + (0.8)(0.70) + 0.5$$

$$\Rightarrow 1.806$$

Output Neuron O_2 :

$$\text{net-}O_2 \Rightarrow w_{35} \cdot h_1 + w_{45} \cdot h_2 + w_{55} \cdot h_3 + b_2$$

$$\Rightarrow (0.3)(0.68) + (0.6)(0.66) + (0.2)(0.70) + 0.5$$

$$\Rightarrow 1.242$$

4. Output Layer Activations:

$$O_1 = \frac{1}{1+e^{-1.806}} \Rightarrow 0.858$$

$$O_2 = \frac{1}{1+e^{-1.242}} \Rightarrow 0.776$$

STEP 2

Error Calculation:-

$$E_{\text{Total}} = E_1 + E_2$$

Error at Output O_1 :

$$E_1 = \left(\frac{1}{2}\right) (t_1 - O_1)^2$$

$$\Rightarrow \left(\frac{1}{2}\right) (1 - 0.858)^2$$

$$\Rightarrow 0.0099$$

Error at Output O_2 :

$$E_2 = \left(\frac{1}{2}\right) (t_2 - O_2)^2$$

$$\Rightarrow \left(\frac{1}{2}\right) (0 - 0.776)^2$$

$$\Rightarrow 0.3011$$

Total Error:

$$E_{\text{Total}} \Rightarrow E_1 + E_2$$

$$\Rightarrow 0.311085$$

STEP 03

BACK PROPAGATION (CHAIN RULE)

$$\frac{\partial E}{\partial w} = \frac{\partial E}{\partial O} \cdot \frac{\partial O}{\partial \text{net}} \cdot \frac{\partial \text{net}}{\partial w}$$

1. Output layer - Chain Rule for O_1 :

We compute three terms using Chain rule:

$$\text{Term 1: } \frac{\partial E_{\text{Total}}}{\partial O_1} \Rightarrow \frac{\partial \left(\frac{1}{2} (t_1 - O_1)^2\right)}{\partial O_1} \Rightarrow -(t_1 - O_1)$$

$$\Rightarrow -(1 - 0.858) \Rightarrow -0.141$$

$$\text{Term 2: } \frac{\partial O_1}{\partial \text{net} - O_1} \Rightarrow \sigma'(\text{net} - O_1) = O_1 (1 - O_1)$$

$$\Rightarrow 0.858 \times (1 - 0.858)$$

$$\Rightarrow 0.121$$

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Output Error Signal (delta) for O_1 :

$$\delta_{O_1} = \frac{\partial \epsilon}{\partial O_1} \cdot \frac{\partial O_1}{\partial \text{net-}O_1}$$

$$\Rightarrow (- (t_1 - O_1)) \cdot (1 - O_1) O_1$$

$$\Rightarrow -0.141 \times 0.121 = -0.017$$

2. Output Layer - Chain Rule for O_2 :

$$\frac{\partial \epsilon_{\text{total}}}{\partial O_2} \Rightarrow (-t_2 - O_2) = (-0 - 0.776)$$

$$\Rightarrow 0.776$$

$$\frac{\partial O_2}{\partial \text{net-}O_2} \Rightarrow O_2 (1 - O_2)$$

$$\Rightarrow (0.776) \times (0.2239)$$

$$\Rightarrow 0.1737$$

$$\delta O_2 \Rightarrow 0.776 \times 0.1737$$

$$\Rightarrow 0.1348$$

3. Gradients for Hidden $\rightarrow$ Output weights:Weights to O_1 :

$$\frac{\partial \epsilon}{\partial w_{34}} \Rightarrow \delta_{O_1} \cdot h_1$$

$$\Rightarrow -0.017 \times 0.681 \Rightarrow -0.0116$$

$$\frac{\partial \epsilon}{\partial w_{44}} \Rightarrow \delta_{O_1} \cdot h_2$$

$$\Rightarrow -0.017 \times 0.66 \Rightarrow -0.0113$$

$$\frac{\partial \epsilon}{\partial w_{54}} \Rightarrow \delta_{O_1} \cdot h_3$$

$$\Rightarrow -0.017 \times 0.76 \Rightarrow -0.0129$$

Weights to O_2 :

$$\frac{\partial \varepsilon}{\partial w_{35}} \Rightarrow \delta - O_2 \cdot h_1 = 0.1348 \times 0.68 \Rightarrow 0.092$$

$$\frac{\partial \varepsilon}{\partial w_{45}} \Rightarrow \delta - O_2 \cdot h_2 \Rightarrow 0.1348 \times 0.66 \Rightarrow 0.089$$

$$\frac{\partial \varepsilon}{\partial w_{55}} \Rightarrow \delta - O_2 \cdot h_3 \Rightarrow 0.1348 \times 0.70 \Rightarrow 0.095$$

Shared Output Bias b_2 :-

$$\begin{aligned} \frac{\partial \varepsilon}{\partial b_2} &\Rightarrow \delta - O_1 + \delta - O_2 \\ &\Rightarrow -0.0170 + 0.1348 \\ &\Rightarrow 0.1177 \end{aligned}$$

4. Hidden Layer — BackPropagating Error

Error Propagated to h_1 :-

$$\begin{aligned} \frac{\partial \varepsilon}{\partial h_1} &\Rightarrow \delta - O_1 \cdot w_{34} + \delta - O_2 \cdot w_{35} \\ &\Rightarrow (-0.0170)(0.7) + (0.1348)(0.3) \\ &\Rightarrow 0.028 \end{aligned}$$

Error Propagated to h_2 :-

$$\begin{aligned} \frac{\partial \varepsilon}{\partial h_2} &\Rightarrow \delta - O_1 \cdot w_{44} + \delta - O_2 \cdot w_{45} \\ &\Rightarrow (-0.0170)(0.4) + (0.1348)(0.6) \\ &\Rightarrow 0.074 \end{aligned}$$

Error Propagated to h_3 :-

$$\begin{aligned} \frac{\partial \varepsilon}{\partial h_3} &\Rightarrow \delta - O_1 \cdot w_{54} + \delta - O_2 \cdot w_{55} \\ &\Rightarrow (-0.0170)(0.8) + (0.1348)(0.2) \\ &\Rightarrow 0.133 \end{aligned}$$

5. Hidden Layer Delta:-

$$\underline{\partial h_1} \Rightarrow h_1(1-h_1)$$

$$\partial_{\text{net}-h_1} \Rightarrow 0.68 \times 0.316 \Rightarrow 0.216$$

$$\delta-h_1 \Rightarrow 0.0285 \times 0.2163 \Rightarrow 0.0061$$

$$\underline{\partial h_2} \Rightarrow h_2(1-h_2)$$

$$\partial_{\text{net}-h_2} \Rightarrow 0.66 \times 0.338 \Rightarrow 0.224$$

$$\delta-h_2 \Rightarrow 0.074 \times 0.224 \Rightarrow 0.0165$$

$$\underline{\partial h_3} \Rightarrow h_3(1-h_3)$$

$$\partial_{\text{net}-h_3} \Rightarrow 0.70 \times 0.295 \Rightarrow 0.208$$

$$\delta-h_3 \Rightarrow 0.133 \times 0.208 \Rightarrow 0.003$$

6. Gradients for Input $\rightarrow$ Hidden weights:-

Weights to h_1 :-

$$\underline{\partial \varepsilon} \Rightarrow \delta-h_1 \cdot x_1$$

$$\frac{\partial \varepsilon}{\partial w_{13}} \Rightarrow 0.0062 \times 0.6 \Rightarrow 0.0034$$

$$\frac{\partial \varepsilon}{\partial w_{23}} \Rightarrow \delta-h_1 \cdot x_2 \Rightarrow 0.0062 \times 0.1 = 0.0006$$

Weights to h_2 :-

$$\underline{\partial \varepsilon} \Rightarrow \delta-h_2 \cdot x_1 \Rightarrow 0.0165 \times 0.6 = 0.0099$$

$$\frac{\partial \varepsilon}{\partial w_{14}}$$

$$\underline{\partial \varepsilon} \Rightarrow \delta-h_2 \cdot x_2 \Rightarrow 0.0165 \times 0.1 = 0.0016$$

$$\frac{\partial \varepsilon}{\partial w_{24}}$$

Weights to h_3 :-

$$\underline{\partial \varepsilon} \Rightarrow \delta-h_3 \cdot x_1 \Rightarrow 0.0027 \times 0.6 \Rightarrow 0.0016$$

$$\frac{\partial \varepsilon}{\partial w_{15}}$$

$$\underline{\partial \varepsilon} \Rightarrow \delta-h_3 \cdot x_2 \Rightarrow 0.0027 \times 0.1 \Rightarrow 0.00027$$

$$\frac{\partial \varepsilon}{\partial w_{25}}$$

Shared Hidden Bias b_1 :-

$$\frac{\partial \mathcal{E}}{\partial b_1} \Rightarrow \delta_{-h_1} + \delta_{-h_2} + \delta_{-h_3}$$

$$\Rightarrow 0.0062 + 0.0165 + 0.0027$$

$$\Rightarrow 0.0255$$

STEP 04

Weight Updates:-

Update Rule:

$$W_{\text{new}} = W_{\text{old}} - \eta \cdot \frac{\partial \mathcal{E}}{\partial w}$$

1. Update Output Weights:-

$$W_{34\text{-new}} \Rightarrow 0.7 - 0.1 \times (-0.0116)$$

$$\Rightarrow 0.701$$

$$W_{44\text{-new}} \Rightarrow 0.4 - 0.1 \times (-0.0113)$$

$$\Rightarrow 0.401$$

$$W_{54\text{-new}} \Rightarrow 0.8 - 0.1 \times (-0.0126)$$

$$\Rightarrow 0.801$$

$$W_{35\text{-new}} \Rightarrow 0.3 - 0.1 \times 0.0921$$

$$\Rightarrow 0.290$$

$$W_{45\text{-new}} \Rightarrow 0.6 - 0.1 \times 0.0892$$

$$\Rightarrow 0.591$$

$$W_{55\text{-new}} \Rightarrow 0.2 - 0.1 \times (0.0950)$$

$$\Rightarrow 0.190$$

$$b_{2\text{-new}} \Rightarrow 0.5 - 0.1 \times 0.1177$$

$$\Rightarrow 0.488$$

2. Update Hidden Weights:-

$$W_{13\text{-new}} = 0.4 - 0.1 \times 0.0037 \Rightarrow 0.399$$

$$W_{23\text{-new}} \Rightarrow 0.3 - 0.1 \times 0.0006 \Rightarrow 0.299$$

$$W_{14\text{-new}} \Rightarrow 0.2 - 0.1 \times 0.0099 \Rightarrow 0.199$$

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$$W_{24\text{-new}} \Rightarrow 0.5 - 0.1 \times 0.00165 \Rightarrow 0.499$$

$$W_{15\text{-new}} \Rightarrow 0.6 - 0.1 \times 0.00166 \Rightarrow 0.599$$

$$W_{25\text{-new}} \Rightarrow 0.1 - 0.1 \times 0.00027 \Rightarrow 0.099$$

$$b_{1\text{-new}} \Rightarrow 0.5 - 0.1 \times 0.0255 \Rightarrow 0.497$$